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ADMM Based Algorithm for Spike and Smooth Signal Separation using Over-Complete Dictionary

K. Harikumar, S. Athira, Yedu C. Nair, V. Sowmya, K.P.Soman.

Abstract— **In signal processing, many a time people deal with smooth stationary signals mixed with sharp spikes and most of the time their analysis demands separation of the smooth and spike elements. In this paper, we propose a methodology for this kind of separation based on the well-known notion of using an over complete dictionary to define an underdetermined system of linear equations and picking out its sparsest solution. The Alternating Direction Method of Multipliers (ADMM) framework is proposed for formulating and solving this optimization problem. The study of the performance of the algorithm with respect to certain signal parameters is also included. Performance of the algorithm was tested for different sparsity values at different signal energies and the results are reported. Cases involving group sparsity is also studied.**

Index Terms—**ADMM, optimization, over-complete dictionary, smooth, sparsity, spike.**

I. INTRODUCTION

THE separation of the signal into its smooth and sharp variations facilitates enhanced signal analysis in many practical situations. For instance, spike detection is a significant step for the analysis of neural recording signals as it helps to minimize the rate of transmission; thus permitting usage of more number of recording channels [1]. Spike detection and extraction is also a key component in seismic data analysis for post stack seismic resolution inversion methods [2] used to derive useful information about the stratigraphic and structural properties of the area/object of interest. Spike detection also plays a crucial role in brain machine interface as described in [3], where the neuroscientist are in need of information regarding the neurons responsible for each spike, to develop a map of neural circuitry. In many other applications spikes are considered as noise and requires to be removed other applications spikes are considered as noise and requires to be removed. Biomedical signals are often corrupted with high amplitude sparse noise [4], the noise of this kind is known to distort the frequency spectrum; even the presence of a single spike will be enough to fill up all the frequency coefficients and thus the removal of such spikes is of utmost importance for the analysis of these signals. Another critical issue in applications like power convertors is the false detection of voltage faults due to intervening spikes [5]. However, effective separation of single neuron activity spikes and the local field potentials (low frequency brain signals), corresponding to network activity, is of great importance in the study of brain function [6]. In such cases, there is a need for a methodology of separation wherein extraction of one component does not destroy the features of the other. Conventional methods like low pass filtering can never achieve this [4], and results in high level of spectral contamination [6]. Other traditional methods include the use of median filter [7], [8] to remove spiked noise. Since the concerned signal of interest is of non-stationary nature [9], [10], [11] discusses the use of wavelets [12] for spike detection and removal. The impulse rejection filter discussed in the article [4], suggests a statistical method of analysis for spike removal to avoid interference from spurious frequencies of the abruptly varying signal with the original frequency components.

The theory of optimization provides an appropriate framework for easy formulation of real-world problems into a mathematical structure which can later be exploited for finding its optimal solution [13]. Many optimization based methods have been developed for solving such problems concerning smooth and spiked signals, and the use of over-complete dictionary to achieve such a feat has gained much popularity [14], [15]. Here, a dictionary is defined as a set of vectors B used to span the k dimensional vector space V. If all the vectors in set B are linearly independent and the cardinality of set B, denoted by l, equals k, then the particular dictionary B is called a basis set for V and if $l > k$ then it is an over-complete dictionary for V. Further, notice that the over-complete dictionary can contain linearly dependent subsets [14].

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There exists infinite possible choices of basis sets for the vector space V, and any union of 2 or more such sets will essentially form an over-complete dictionary for the space. Suppose, as in the above problem, we need to separate the signal s into two components s_1 and s_2 which are better expressed using two different basis sets B_1 and B_2, then we can achieve their separation by projecting the signal s onto the over-complete dictionary $B = B_1 \cup B_2$. The projection not being unique is now cast as an optimization problem in such a way that the coefficients corresponding to the basis set $B_1 \subseteq B$ captures all the information of signal s_1 and coefficients corresponding to the basis set $B_2 \subseteq B$ captures all the information of signal s_2. In this case, the notion of best expressed can be taken as the representation of the signals s_1 and s_2 with least number of vectors from set B_1 and B_2 respectively. Here, the optimization routine is designed to look for the sparsest solution for projecting signal s onto B. A choice o

/f orthonormal basis sets satisfying mutual incoherence can be used to form such an over complete dictionary [14], [16]. This optimization problem can be efficiently solved using the method of Alternating Direction Method of Multipliers (ADMM) [17].

ADMM provides the solution of constrained convex optimization problems by converting them into unconstrained optimization problems by the method of Lagrangian multipliers [18]. This method combines the benefits of decomposability of the method of dual ascent with that of the better convergence properties of the method of multipliers. This decomposability is what allows the implementation of the algorithm in distributed frameworks. Hence this method is especially suited for large scale problems in distributed scenarios [17].

The remainder of this paper is organized as follows. Section II describes the problem and provides the proposed problem formulation. Section III discusses the performance of the algorithm with results obtained for various test inputs and finally the conclusion is drawn in Section IV.

II. PROBLEM DESCRIPTION

In this paper we analyze the case of a signal $x \in \mathrm{IR}^N$ which can be considered as a unique sum of two signals $v \in \mathrm{IR}^N$ and $w \in \mathrm{IR}^N$ where v is a k_1 sparse signal with respect to the DCT basis [19], [20] (exactly k_1 non zero DCT coefficients) and w is a k_2 sparse signal with respect to the trivial/identity basis. Our aim through this algorithm is to separate the mixture into these two components. Hence, in this case the over complete dictionary A will be of the form $[D \;\; I]_{2N \times N}$ where D contains the orthonormal DCT basis vectors in its columns and I is the identity matrix. Now x can be represented in terms of this over-complete basis as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_N \end{bmatrix}_{N\times 1} = \begin{bmatrix} D_{N\times N} & I_{N\times N} \end{bmatrix}_{N\times 2N} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ . \\ . \\ \alpha_{2N} \end{bmatrix}_{2N\times 1} \tag{1}$$

where $\alpha = [\alpha_1, \alpha_2, ...\alpha_{2N}]^T$ are the coefficients for the over complete basis. Now this underdetermined system has infinitely many solutions out of which the sparsest solution [14] will have only $(k_1 + k_2)$ non zero elements. Here, the first N components of the solution corresponds to the smooth part of the signal, represented perfectly through k_1 DCT basis, and the last N components is a k_2 sparse vector, representing the spike. Theoretically, for this unique $(k_1 + k_2)$ sparse solution to exist, any set of $2(k_1 + k_2)$ column vectors from A should be linearly independent [21], [14]; in other words the Spark of A should be at least $2(k_1 + k_2) + 1$. This can be formulated as an optimization problem [22] as follows:

$$\begin{aligned} &\min_{\alpha} \|\alpha\|_0 \\ &s.t\ A\alpha = x \end{aligned} \tag{2}$$

The l_0 norm minimization of such a solution set is an NP hard problem [23] and hence is not computationally tractable. But under certain favorable conditions [24], [25] the solution that minimizes the l_1 norm also approaches the sparsest solution; hence the problem can be reformulated as

$$\begin{aligned} &\min_{\alpha} \|\alpha\|_1 \\ &s.t\ A\alpha = x \end{aligned} \tag{3}$$

This typical optimization problem can be reformulated in an ADMM [17] framework, which is a better option for solving massive problems than conventional optimization algorithms and can be expressed as

$$\begin{aligned} &\min_{\alpha} \|\alpha\|_1 + g(z) \\ &s.t\ \alpha - z = 0 \end{aligned} \tag{4}$$

where z is a surrogate variable for α and $g(z)$ is the indicator function for the condition A α = x expressed as

$$g(z) = \begin{cases} 0 & \{z \mid Az = x\} \\ \infty & otherwise \end{cases} \tag{5}$$

Converting this constrained optimization problem into an unconstrained optimization problem by using Lagrangian multiplier eases computations associated with the calculation, and the convergence rate can further be enhanced by adding an augmented Lagrangian term [26]. The problem can now be expressed as

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$$\min \|\alpha\|_1 + g(z) + \rho/2\|\alpha - z\|_2^2 + \lambda^T(\alpha - z) \tag{6}$$

where λ is the Lagrangian multiplier for the constraint and ρ is the multiplier for the augmented parabolic term. Since the functions in the objective are separable, the minimizations over α and z can be done separately. Thus,

$$\alpha^* = \arg\min_{\alpha} \|\alpha\|_1 + \rho/2\|\alpha - z\|_2^2 + \lambda^T(\alpha - z) \tag{7}$$

$$z^* = \arg\min_{z} g(z) + \rho/2\|\alpha - z\|_2^2 + \lambda^T(\alpha - z); \quad dom \quad g = \{z \mid Az = \alpha\}, \tag{8}$$

The update expressions for α and z for the above formulation is then given by

$$\alpha^{k+1} = S_{1/\rho}(z^k - \lambda^k/\rho) \tag{9}$$

where the soft thresholding [27], [17] operator S is defined as

$$S_k(a) = \begin{cases} a - k; & a > k \\ 0; & |a| \le k \\ a + k; & a < -k \end{cases} \tag{10}$$

and the z update is given by the Euclidean projection of $\alpha + \lambda/\rho$ onto the affine space $Az = x$. The projection of a vector onto an affine space [19] can be visualized as in Fig. 1 and is given by

$$z^{k+1} = (\alpha + \frac{\lambda^k}{\rho}) - A^T A^{-1}(\alpha^{k+1} + \frac{\lambda^k}{\rho}) + A^+ x \tag{11}$$

where $A+$ denotes the pseudo-inverse of A. The Lagrangian variable, being the dual variable in terms of the Lagrangian function needs to be updated in the direction of increase of the function and hence,

$$\lambda^{k+1} = \lambda^k + \rho(\alpha^{k+1} - z^{k+1}) \tag{12}$$

The variables are iteratively updated as per the above expressions until satisfactory convergence conditions [17] are met. The signals can be then recovered as follows:

$$v_{rec} = D\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix} \tag{13}$$

$$w_{rec} = \begin{bmatrix} \alpha_{N+1} \\ \alpha_{N+2} \\ \vdots \\ \alpha_{2N} \end{bmatrix} \tag{14}$$

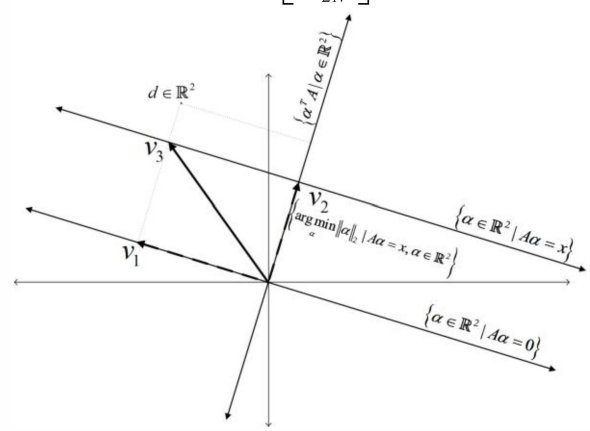

Fig. 1. A is a 2X 2 matrix with rank 1 and x is an element in R^2 belonging to the column space of A. $\{\alpha \in R^2 \mid A\alpha = 0\}$ represents the nullspace, $\{\alpha \in R^2 \mid A\alpha = x\}$ represents the affine space(which is not a vector space) and $\{\alpha^T A \mid \alpha \in R^2\}$ represents the row space. Suppose d is an arbitrary point in R^2 and the task is to find the vector v_3 such that v_3 is the projection of d onto the affine space. v_3 can be expressed as a $v_3 = v_1 + v_2$, where v_1 represents the projection of \$d\$ onto the nullspace and v_2 is the least norm solution to the underdetermined system of linear equation $A\alpha = x$. The above scenario can be extended to any higher dimension.

III. Results and Observations

In the experiments presented in the paper, ratio of powers of smooth signal to that of the spike signal was maintained to be 0 dB; this was so chosen just to avoid any bias between the two components v and w during their separation. Similar separability experiments were done for different power ratios and good separability was observed for power ratios in the range of [-20, 20] dB. Let N_v be a set of k_1 numbers chosen randomly from Z_a = {0, 1, 2...N − 1}, Nw be a set of k_2 numbers chosen randomly from Za and $v' = [v_1' \, v_2' \ldots v_N']^T$ then

$$v_i' = \begin{cases} v_i' & i \in N_v \\ 0 & i \notin N_v \end{cases} \tag{15}$$

$$w_i = \begin{cases} w_i & i \in N_w \\ 0 & i \notin N_w \end{cases} \tag{16}$$

$$\|v'\|_2 + \|w\|_2 = c \tag{17}$$

where

$$v = Dv' \tag{18}$$

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$c \in$ IR. For $i \in N_v$ and $j \in N_w$, the values of v_i' and w_j are taken randomly form a uniform distribution [−.5, .5] and appropriately scaled so as to satisfy Eq. 17. The error factor E_f is defined as

$$E_f = 10 \times \log_{10}\left(\left\| \begin{bmatrix} \frac{\|v - v_{rec}\|_2}{N} & \frac{\|w - w_{rec}\|_2}{N} \end{bmatrix} \right\|_\infty \right) \quad (19)$$

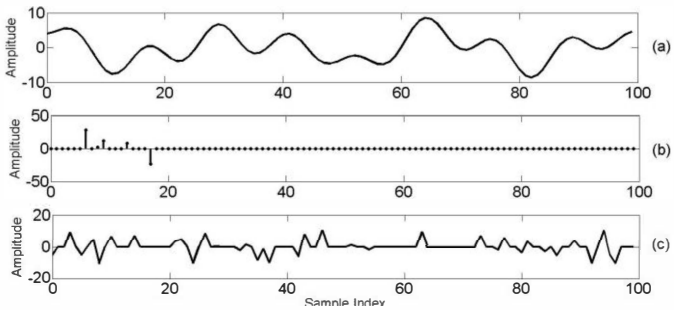

Fig. 2. (a) Smooth signal v (b) DCT of the smooth signal v (c) Spiked signal w.

The separability was considered to be good if the error factor E_f is less than 40dB. Throughout the experiments c was taken to be 40. Fig. 2(a) shows the smooth portion of a test signal that was taken with k_1 = 5 being the sparsity in the DCT domain and the corresponding DCT coefficients is shown in

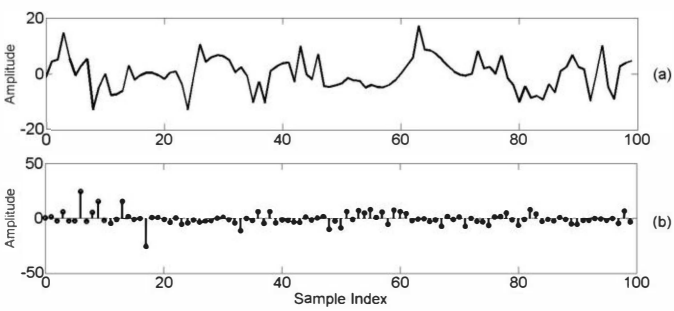

Fig. 3. (a) Composite signal x. (b) DCT coefficients of x; the DCT occupies almost all the coefficients

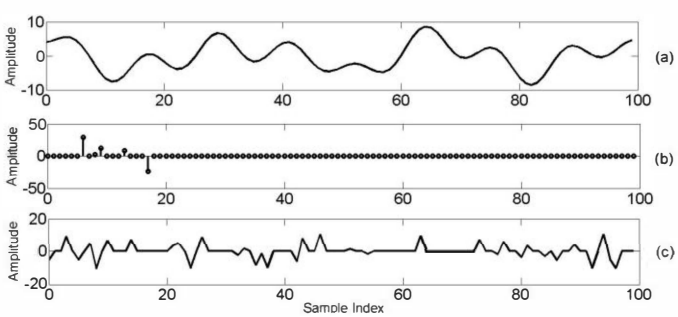

Fig. 4. (a) Reconstructed smooth signal v_{rec}. (b) DCT of the reconstructed smooth signal v'_{rec} (c) Reconstructed spiked signal w_{rec}

Fig. 2(b). The spike signal added to this signal with sparsity k_2= 35 is also shown in Fig. 2(c). The resultant signal after the addition of the spike to the smooth signal and its DCT coefficients are respectively shown in Fig. 3(a) and Fig. 3(b). This signal was applied to the algorithm developed and the results were observed. Fig. 4(a) shows the recovered smooth portion of the signal and Fig. 4(b) the corresponding DCT coefficients. The recovered spike component of the signal is shown in Fig. 4(c) and the corresponding E_f was observed to be -648 dB. Experiments were done in the above setting for all possible combinations of k_1 and k_2, where k_1, $k_2 \in N^+$ and k1 + $k_2 \leq 50$.

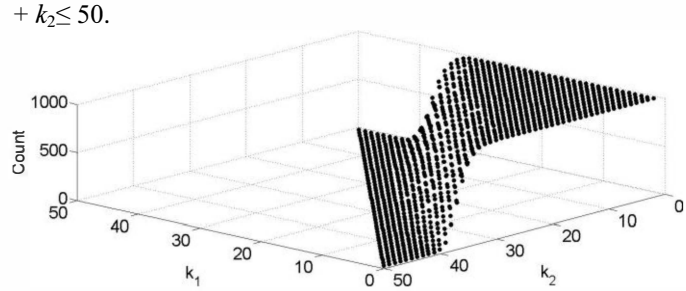

Fig. 5. Count of good separations in 1000 cases versus the sparsity v' and w

Each combination was iterated 1000 times with different signals and a count of good separations were taken as a measure for comparison of the performance of the algorithm for different sparsity values and the results are given in Fig. 5. For a broader picture, the count for good separation was averaged for all possible k_1 and k_2 giving a particular $k = k_1 + k_2$and is shown in Fig. 6. The algorithm shows very good separability for any total sparsity value of k<30 and separability drops as k goes beyond 35. Also, it was observed that for a particular value of k itself, the count typically depends on values of k_1 and k_2; in general it was observed that better separability was achieved in cases where k_2 was less.

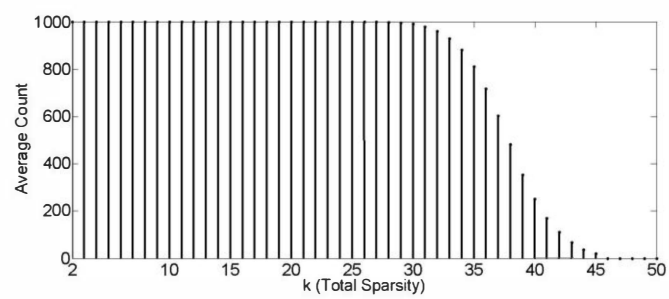

Fig. 6. Average count of good separation versus total sparsity k.

Further experiments showed that the algorithm works acceptably well if the sparse components are distributed over the whole range Z_a and not in clumps or groups. Fig. 8(a) shows another test signal where it takes the same smooth signal as in Fig. 2(a) but the components of spike signal occurs in a cluster as shown in Fig. 7(b).

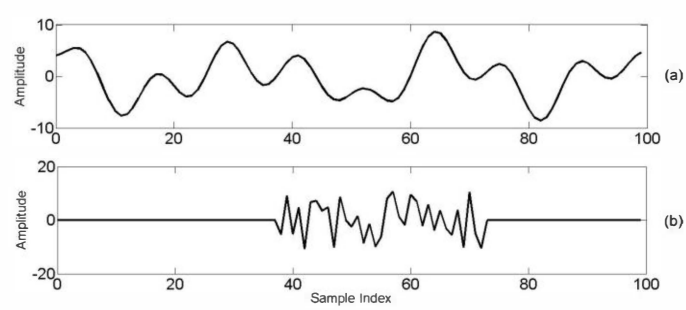

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Fig.7. (a) Smooth signal *v*. (b) Spiked signal *w*.

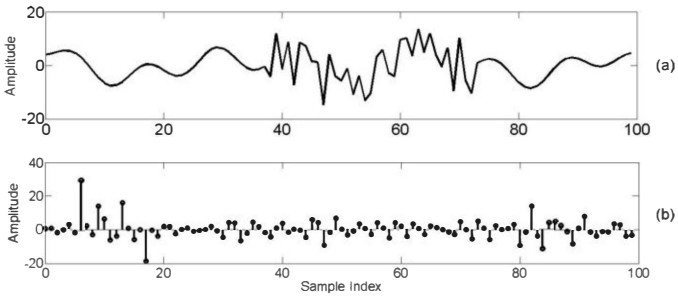

Fig. 8. (a) Composite signal *x*. (b) DCT coefficients of *x*.

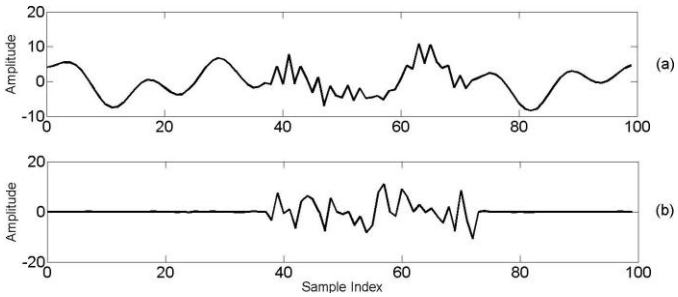

Fig. 9. Reconstructed smooth signal v_{rec}. (b) Reconstructed spiked signal w_{rec}.

The DCT coefficients of this test signal are shown in Fig. 8(b). The reconstructed signals are shown in Fig. 9 and the error factor E_f was observed to be 5.8534dB which considerably high and the reconstructed signal appears distorted. The ADMM implementation of the algorithm was found to be fast and efficient. Further the framework allows easy scaling up of the data since this methodology is especially suited for large data sets.

IV. Conclusion

The proposed algorithm is capable of efficient separation of smooth and spike components of a signal for all cases where the ratio of power of smooth signal to the power of spike signal was in the range of -20dB to +20dB. The smoothness and spike nature can be extended to a maximum sparsity level of $k \leq 35$ for a signal of length 100. Further the ADMM algorithm proposed for the problem was observed to be quite efficient.

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